

PatternShop — Capstone Grading Rubric

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100 points · Brief: spec.en.md · Türkçe: rubric.tr.md

1 How grading works

- The rubric has **10 criteria in three groups**. The groups match the syllabus split of the capstone's 40 % of the course grade: specification and design 25 points (10 %), implementation and tests 50 points (20 %), report and defence 25 points (10 %).
- Every criterion has **four levels**: Excellent, Good, Adequate, Insufficient. Each level is defined by things a grader can check in the submission — a count, a command, a file, a section. The grader picks the highest level whose conditions are **all** met and gives exactly the points of that level.
- Insufficient gives its points only if the deliverable exists; a missing deliverable scores 0.
- The graded state is the commit tagged `capstone-final` in the student's fork (W14, before the presentation slot). `SPEC.md` is additionally graded in the version submitted at the W10 milestone (C1).
- Graders run, on JDK 27, from the repository root:
`./mvnw -q -pl capstone/starter verify, ./mvnw -q -pl capstone/starter test -Pexercises`
and `./mvnw -q -pl capstone/starter verify -Pcoverage`.

2 Automatic fail conditions

If **any** of the following holds on the tagged commit, the implementation part fails: **criteria C3–C8 score 0** (50 points). C1, C2, C9 and C10 are still graded.

Gate	Condition	How it is checked
G1	The build fails on JDK 27: compilation error, a <code>-Xlint:all -Werror</code> warning, or a red test of the student's own	<code>./mvnw -q -pl capstone/starter verify</code>
G2	Any of the 83 given acceptance tests is red (failure, error or not run)	<code>./mvnw -q -pl capstone/starter test -Pexercises</code>
G3	Any of the 7 given architecture rules is red	ShopArchitectureTest in the verify run
G4	A given test, fixture, test resource, architecture rule or GIVEN API type was modified, deleted, disabled (<code>@Disabled</code> , removed tag, Surefire exclusion) or weakened	<code>git diff</code> against the published starter for <code>capstone/starter/src/test/**/acceptance/**</code> , the <code>*ExerciseTest</code> and ShopArchitectureTest bindings, <code>src/test/resources/acceptance/**</code> and <code>.../capstone/api/**</code>

The only allowed build change is the one extension E10 needs (`--enable-preview` isolated as in m10); all given tests must still pass without preview. Academic-integrity violations are handled separately (brief §12).

3 Criteria and weights

Group	Id	Criterion	Points
Specification and design	C1	Student SPEC.md	15
	C2	Pattern plan and justification table	10
Implementation and tests	C3	Correctness beyond the gate (extension features)	10
	C4	Pattern use	14
	C5	Modern Java and concurrency	8
	C6	Architecture	6
	C7	Test quality	7
	C8	Code quality and process	5
	C9	Report	12
Report and defence	C10	Presentation and defence	13
	Total		100

4 Level descriptors

4.1 C1 — Student SPEC.md (15)

Checklist (W10 version for a–g, final version for h): (a) all ten template sections present; (b) every mandatory feature F1–F11 mapped to its acceptance suite; (c) each extension feature has at least three Given / When / Then acceptance criteria and named test classes; (d) a Mermaid domain class diagram and a hexagon diagram; (e) a pattern plan with at least ten rows; (f) at least three boundaries or assumptions where the brief is silent; (g) a week-by-week plan for W10–W13; (h) the final SPEC.md matches the code — every planned pattern and extension exists, or the change log says why it changed.

Level	Points	Descriptor
Excellent	15	Submitted by the W10 deadline; all of a–h
Good	11	Submitted by the W10 deadline; six or seven of a–h
Adequate	7	Four or five of a–h, or submitted up to one week late
Insufficient	3	Three or fewer of a–h, or submitted more than one week late

4.2 C2 — Pattern plan and justification table (10)

A row is **complete** when it names (1) a force or problem specific to PatternShop, (2) the participant types, which exist in the code, (3) an alternative considered and why it was not chosen, and (4) a test that shows the pattern working. Table template: section 5.

Level	Points	Descriptor
Excellent	10	At least 10 complete rows, and at least one pattern deliberately <i>not</i> used with a reason
Good	7	At least 10 rows, at least 8 of them complete
Adequate	5	At least 10 rows with at least 5 complete, or 8–9 rows all complete
Insufficient	2	Fewer than 8 complete rows

4.3 C3 — Correctness beyond the gate (10)

The gate (G2) already requires all given acceptance tests to pass. C3 grades the **extension features** (two; pairs three). An extension is **complete** when every acceptance criterion written for it in `SPEC.md` has at least one green test that names it (in the test name or `@DisplayName`) and the feature is reachable from the CLI.

Level	Points	Descriptor
Excellent	10	All extensions complete
Good	7	All extensions implemented; at least 75 % of their criteria have green tests
Adequate	5	One extension complete, or all extensions with at least 50 % of their criteria tested green
Insufficient	2	No extension reaches 50 %

4.4 C4 — Pattern use (14)

For every pattern declared with `@PatternRole` and counted (creational, structural, behavioural, concurrency), the grader checks: the annotated types play the roles the module lesson gives for that pattern, and the pattern runs on a path covered by a green acceptance or unit test. A pattern meeting both is **valid**. The **mix** is at least 2 creational, 2 structural, 3 behavioural and 1 concurrency pattern (not Immutable Object).

Level	Points	Descriptor
Excellent	14	At least 12 valid patterns, mix met by valid patterns, and no declared pattern is invalid
Good	10	10 or 11 valid patterns, mix met by valid patterns
Adequate	7	8 or 9 valid patterns, or at least 10 valid patterns without the mix
Insufficient	3	7 or fewer valid patterns

4.5 C5 — Modern Java and concurrency (8)

Checklist: (a) at least one sealed hierarchy of records handled with an exhaustive `switch` with record patterns and no `default`; (b) at least two further sealed hierarchies or exhaustive switches in the student's own code; (c) all value types are records (no mutable value classes); (d) single-method strategies or commands are lambdas or method references where no state is needed; (e) fulfilment runs on virtual threads with a bounded number of parallel orders, and `SPEC.md` §6 argues why shared state is thread-safe; (f) `Optional` only as a return type; no public method returns `null`.

Level	Points	Descriptor
Excellent	8	All of a–f
Good	6	Five of a–f
Adequate	4	Three or four of a–f, including (a) and (e)
Insufficient	2	Otherwise

4.6 C6 — Architecture (6)

The gate (G3) already requires the given rules to be green. Checklist: (a) adapters are constructed only in the composition root (`config`); (b) outbound ports are interfaces owned by `application` and named after their role, not a technology; (c) business outcomes are sealed results, infrastructure failures are exceptions; (d) at least one additional ArchUnit rule of the student's own, green and meaningful (for example: ports are interfaces, naming of adapters); (e) the hexagon diagram in `SPEC.md` matches the actual packages.

Level	Points	Descriptor
Excellent	6	All of a–e
Good	4	Four of a–e
Adequate	3	Three of a–e
Insufficient	1	Two or fewer of a–e

4.7 C7 — Test quality (7)

Checklist: (a) every valid pattern (C4) has at least one own unit test that shows its behaviour; (b) test doubles are hand-written; no mocking library; (c) test names describe behaviour (`appliesCouponAfterThresholdDiscount`, not `test1`); (d) tests are deterministic: no `Thread.sleep` for synchronisation, time comes from an injected clock; (e) line coverage of `domain` and `application` is at least 80 % (JaCoCo report of `-Pcoverage`); (f) every extension test is listed next to its acceptance criterion in `SPEC.md`.

Level	Points	Descriptor
Excellent	7	All of a–f
Good	5	Five of a–f
Adequate	3	Three or four of a–f
Insufficient	1	Two or fewer of a–f

4.8 C8 — Code quality and process (5)

Checklist: (a) no `@SuppressWarnings` without a comment explaining why; (b) every public type in `domain` and `application` has a one-line Javadoc of intent; (c) no class has more than 10 public methods and no

method is longer than 30 lines; (d) at least one commit in each of the weeks W9–W13, with Conventional Commit messages; (e) no leftover `TODO(capstone)` markers, no commented-out code, no unused classes.

Level	Points	Descriptor
Excellent	5	All of a–e
Good	4	Four of a–e
Adequate	2	Three of a–e
Insufficient	1	Two or fewer of a–e

4.9 C9 — Report (12)

Checklist for `REPORT.md`: (a) a design overview with the hexagon diagram; (b) the complete pattern-justification table (may be the same as in `SPEC.md`, updated); (c) at least two modern-Java decisions explained against the classic alternative; (d) the concurrency design with its thread-safety argument; (e) a reflection: at least two things you would change, and one pattern you deliberately did not use and why; (f) an AI-usage statement (what for, which parts); (g) a one-page summary in the other language (English ↔ Turkish); (h) 6–10 pages, terminology as in `docs/glossary.md`, every claim about the code points to a file path.

Level	Points	Descriptor
Excellent	12	All of a–h
Good	9	Six or seven of a–h
Adequate	6	Four or five of a–h
Insufficient	2	Three or fewer of a–h

4.10 C10 — Presentation and defence (13)

Checklist: (a) the talk takes 9–11 minutes; (b) a live CLI demo of checkout, fulfilment and one report works; (c) three patterns are explained with their force, structure and the student's code; (d) one trade-off is presented with the alternative that was rejected. Defence: the examiner asks four questions about the student's own code — locate the participants of a named pattern, explain what a named test proves, sketch how a given change request would be implemented, explain why fulfilment is thread-safe. Pairs: each member presents a part and answers at least two of the questions.

Level	Points	Descriptor
Excellent	13	All of a–d, and four of four questions answered correctly
Good	10	At least three of a–d including (b), and three of four questions answered correctly
Adequate	6	At least two of a–d, and two of four questions answered correctly
Insufficient	3	Otherwise (presentation given)

5 Pattern-justification table template

Copy this table into `SPEC.md` (§5) and `REPORT.md`. One row per counted pattern; add rows for architectural patterns if you like (they do not count towards the ten). The first row is an example of the expected depth.

#	Pattern (category)	Force / problem in Pattern-Shop	Participants (types and @PatternFole roles)	Alternative considered and why not	Modern Java form	Test(s) that show it	Course module
1	Strategy (behavioural)	Four promotion kinds must be priced the same way, and new kinds will be added without touching the pricing pipeline	PromotionRule (strategy), BuyXGetYFreeRule, CategoryPercentOffRule, ... (concrete strategies), PricingPipeline (context)	switch over the promotion spec inside the pipeline: shorter today, but every new kind changes the pipeline and its tests	sealed interface of records; lambdas not used because each rule carries data	PricingAcceptance.categoryPercentOffRoundHalfUpPerLine, PromotionRuleTest.buyXGetYCountsWholeGroupsOnly	m06
2							
...							

Patterns considered and rejected (at least one for the Excellent level in C2):

Pattern	Where it was tempting	Why it was not used
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6 Grading sheet

Id	Criterion	Max	Level	Points	Evidence	(files, commands, notes)
G1–G4	Gates passed? (yes / no — if no, C3–C8 = 0)	—				
C1	Student SPEC.md	15				
C2	Pattern plan and justification table	10				
C3	Correctness beyond the gate	10				
C4	Pattern use	14				
C5	Modern Java and concurrency	8				
C6	Architecture	6				
C7	Test quality	7				
C8	Code quality and process	5				
C9	Report	12				
C10	Presentation and defence	13				
	Total	100				